

Bayesian Network-Based Estimation and Comparison of Player Skill Over Time

Category: Theoretical_AI

CSE 516 — Probabilistic Graphical Models | Major Project Proposal (M1)

1. Title & Category

Title: Bayesian Network-Based Estimation and Comparison of Player Skill Over Time.

Category: Theoretical AI. We study one model in depth and compare different ways of doing inference (the step where hidden skill values are calculated) on it. We are not building a domain-specific application — our goal is to deeply understand, validate, and test one PGM method against clear baselines.

2. Project Summary

Rating systems like Elo and TrueSkill try to guess how skilled a player is, using only the results of past matches (who won or lost). The problem is that most of these systems only look backward in time — they update a player's skill after every match, but never go back and correct earlier guesses using later information. This means a new player's very first rating is usually wrong and stays unreliable for a while, and it also becomes hard to fairly compare players from very different time periods. The TrueSkill Through Time (TTT) model addresses this limitation by connecting every match, of every player, across all of history, into a single Bayesian network. Instead of only moving information forward in time, this network lets information flow both backward and forward, so early estimates become accurate too.

Our mathematical question is: how much does allowing information to flow in both directions actually improve skill estimates, and what is the computational cost of that improvement? We assume performances are approximately Gaussian around a player's true skill and that skill changes slowly and smoothly over time. In this project, we will rebuild this model ourselves, clearly separate how it represents skill (Representation), how it learns its few settings from data (Learning), and how it calculates the final skill values (Inference). We will compare three inference methods on this network, test everything on a large real dataset of tennis matches, and benchmark our results against the original (forward-only) TrueSkill system from Microsoft Research and a newer Bayesian method from Google DeepMind.

3. Key Objectives

1. Draw and explain the network: show how a player's skill today depends on their skill yesterday, and how skill connects to actual match results, using a Bayesian network, and state the assumptions this relies on (Gaussian performance, smooth skill change over time).
2. Convert this network into a form a computer can run calculations on (a factor graph), and implement three inference methods: an exact method for simple two-team matches, an approximate method for matches with more than two teams, and a method that correctly handles win/lose/draw outcomes.
3. Measure how much better the 'looks both forward and backward' approach (TTT) is compared to the older 'only looks forward' baseline (original TrueSkill) and a simple Elo baseline, using clear numbers, not just claims.
4. Learn the model's few settings (such as how much a player's skill can change over time) directly from data instead of guessing them, and test how stable the results are when we change the amount of data, add noise, or remove information.

5. Compare our results with two real published works — the original TrueSkill paper from Microsoft Research (2006) and a recent Bayesian inference method from Google DeepMind (2024) — and clearly state where our project stands next to them.

4. Expected Outcomes

- Our own working code that builds the network and runs all three skill-calculation methods, written and understood by us (not just a copy of an existing package).
- A results table comparing four methods — a simple constant-skill model, Elo, the original (forward-only) TrueSkill, and TTT — using measures such as prediction accuracy, how well-calibrated the win probabilities are, and log-likelihood, each reported as an average across multiple runs with standard deviation, not a single number.
- A graph showing the trade-off between accuracy and computation time for the different inference methods, plus how many rounds of calculation are needed before results stop changing (convergence).
- A robustness study showing how results change with less data, added noise, or different settings, including a basic significance test (e.g., a paired test) to check whether an improvement is real or just chance.
- A short, clear written comparison explaining when this kind of Bayesian network approach is worth its extra computation, and when a simpler method is good enough — directly answering our mathematical question from Section 2.

5. Dataset(s)

Primary dataset: ATP (Association of Tennis Professionals) historical match records, originally collected by Jeff Sackmann and packaged for this type of research by Landfried.

- Size: about 447,000 singles and doubles matches played between 1915 and 2020, involving more than 19,000 players.
- What it contains: for each match — the players involved, whether it was singles or doubles, the tournament name and start date, the round of the tournament, and the playing surface (clay, grass, hard court).
- Why it fits our project: matches are naturally ordered by date, which is exactly what we need to build the 'skill changes over time' part of the network. The dataset is also large enough to properly test whether our inference methods scale and converge correctly.
- Official replication package: the exact data and code used to produce the base paper's published results are also available directly from the Journal of Statistical Software, alongside the R package, at <https://www.jstatsoft.org/article/view/v112i06>. We will use this as our reference to confirm our own implementation reproduces the paper's setup correctly before extending it.
- We will also generate a small synthetic dataset with a known, made-up 'true' skill trajectory for a fake player. This lets us check that our model's estimates are actually correct before trusting it on the real tennis data, which supports the mathematical validation expected in Theoretical AI projects.

6. GitHub Repository

https://github.com/veera-bharakhada/g3_th_apl_gaming-rating_team

7. References

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